

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Coursework:** Data Structures & Algorithms, OOP, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

MindMosaic

Software Engineering Intern

Aug 2025 – May 2026

East Lansing, MI

- Built and shipped an interactive knowledge-graph platform that keeps dense, interconnected content navigable as it grows, delivering a production MVP to real users on a polyglot Java Spring Boot, Neo4j, and PostgreSQL stack.
- Migrated hot read paths off recursive SQL self-joins onto a Neo4j graph layer so relationships stay first-class, cutting query latency 48% (850ms to 440ms) and making multi-hop traversals 3x faster than the equivalent SQL.
- Owned the Spring Boot backend serving the platform's core CRUD and graph-traversal APIs, structuring endpoints around the dual-store model so the client never has to know which database backs a given read.
- Authored an AWS deployment script and managed the environment as code with Terraform, replacing a manual 30-minute multi-service bring-up with a single-command, version-controlled provision.

Claude Builder Club @ MSU

Co-Founder & Vice President

Oct 2025 – Present

East Lansing, MI

- Co-founded and grew an MSU engineering community from zero to 150+ members, replacing a sprawl of spreadsheets, Slack threads, and manual invites with a single club-operations platform I designed and built.
- Built event check-in as a single atomic database RPC that validates a QR token, records attendance, and awards points server-side, making door-time check-ins concurrency-safe and idempotent, sustaining 150+ check-ins per event.
- Shipped the platform continuously from main on a GitHub Actions pipeline that applies database migrations and deploys edge functions on every push, taking the team from manual changes to reviewed continuous deploys.

PROJECTS

Dadei | *React, FastAPI, Redis*

dadei.app

Ambient voice assistant that turns overheard conversation into human-approved calendar, email, and task actions

- Built a React client that streams microphone audio up over a WebSocket with client-side Web Audio processing, receiving incremental live captions back in under 300ms as the backend transcribes each utterance.
- Decoupled background workers from the live FastAPI service behind a Redis Streams event bus with consumer groups, dead-letter routing after 5 retries, and stale-claim recovery, giving at-least-once delivery that pub/sub dropped.

fliks | *Go, PostgreSQL, AWS (EC2, S3)*

github.com/fliks-gg

Multi-game platform that crowd-validates gameplay skill and mints shareable certification cards

- Built a self-hosted video transcoding pipeline with a concurrent Go worker that pulls upload jobs, transcodes, and writes processed video back to object storage, owning the media path end to end instead of renting a managed service.
- Coordinated transcode jobs through a Postgres-backed queue using SELECT FOR UPDATE SKIP LOCKED instead of bolting on Redis or SQS, keeping the whole system on one datastore and one self-provisioned EC2 box.

WizViz

devpost.com/software/wizviz

Webcam fighting game driven by real-time body-gesture recognition, winner of the Interactive Media track at SpartaHack X

- Ran pose inference asynchronously off the render loop with a monotonic timestamp guard that drops out-of-order results, sustaining 60 FPS with sub-20ms input lag instead of blocking on per-frame detection.

TECHNICAL SKILLS

Languages Java, JavaScript/TypeScript, Python, SQL, Go
Frameworks Spring Boot, React, FastAPI
Databases PostgreSQL, Redis, Neo4j
Cloud & Infrastructure AWS (EC2, S3, RDS), Terraform, CI/CD
Tools Git, Linux